

Santander Bank



Know Which Customers Are Leaving - **Before They Do.**

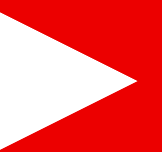
160 million customers, across 40+ countries.

~6.4 million are quietly disengaging.

(Source: Santander Annual Report 2023)

We built a system to find them in time to act.

Prepared for Santander Senior Management



Meet The Team



Bhavisha Chafekar
MSBA



Parul Chaudhary
MSBA



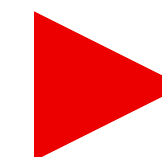
Shivanshu Dagur
MSBA

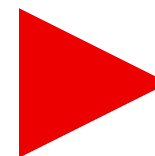


Madhu Damani
MSBA



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MSBA





Built on Real Santander Data

76,020 Real Customer Profiles



76,020

Training
Customer Records



369

Measurable Signals
Per Customer



3.9%

Dissatisfied
(Minority Class)



24:1

Class Imbalance
Ratio

A model that predicts "satisfied" for every customer achieves 96.1% accuracy – but catches zero at-risk customers. We optimised for a different metric: AUC-ROC, which measures how well the model separates the two groups regardless of imbalance.

The Business Problem

Lost Customers Give no warning

“Customers don’t complain, they just leave”



\$300–\$500

Average annual revenue per retail customer.



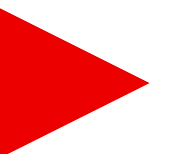
5–7x

More expensive to acquire than retain a customer

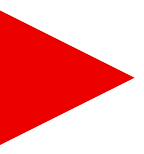


\$256M

Revenue protected if just 10% of at-risk are saved



The Question We Set Out to Answer



CORE QUESTION

"Can we identify dissatisfied customers *early enough to act* — before they leave?"

01 · WHO

Who is at risk, and what behaviours signal it?

LEADS TO →

Key Risk Signals

02 · HOW WELL

How accurately can we predict it?

LEADS TO →

Prediction Accuracy

03 · SO WHAT

What is the business value of acting on this?

LEADS TO →

Revenue at Stake

From Raw Data to Risk Signals

HOW WE TURNED 369 COLUMNS INTO 3 RISK SIGNALS



Age threshold

9%

satisfied under 23

Age rules

Age bins

Age * wealth

Account silence

4 to 5×

higher risk
inactive

Zero count

Log scaling

Balance trends

3×

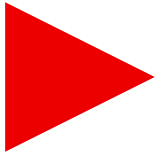
more likely to leave

Balance delta

Total balance

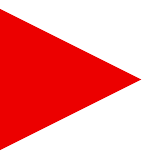
One risk score per customer

Ranked 0 to 1 · Higher score = more at risk of leaving



What Drives the Risk Score

Five Signals Do Most of the Work



Customer Age

var15



26.8%

Cash Balance

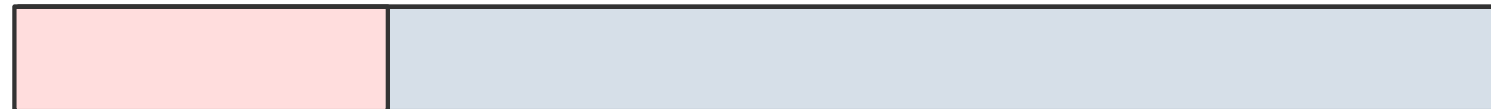
saldo_var30



16.8%

Net Worth / Mortgage

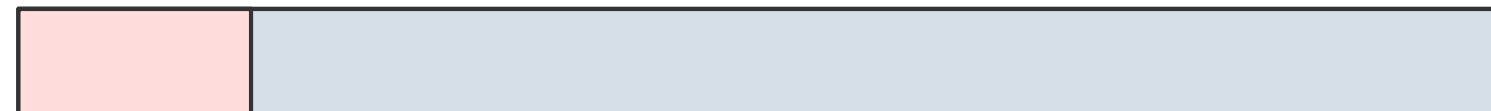
var38



6.9%

Account Inactivity

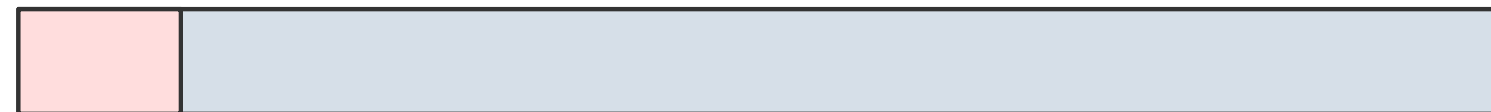
count_zeros (engineered)



4.3%

Balance 3 Months Ago

saldo_medio_var5_hace3



3%

Age, current balance, net worth, account inactivity, and a recent balance trend — five things any relationship manager already wants to know.

Six Independent Views, One Reliable Score

How We Built Confidence in the Answer

STEP 01



Understand the Data

Study customer records, clean errors, and find the patterns that separate happy from at-risk customers.

STEP 02



Enrich the Signals

Create smarter inputs like recent balance changes and total wallet size, that reveal behavior raw data misses.

STEP 03



Train Multiple Models

We combine six optimized approaches (i.e. LightGBM, XGBoost, CatBoost, Random Forest, a Neural Network, and Logistic Regression) into one high-performing system.

STEP 04



Combine the Best

We then blend their outputs into a single, unified prediction, like consolidating multiple expert judgments into one clear, more reliable decision.



Why six models?

Each model has different blind spots. Combining them cancels out weaknesses like a panel of experts voting together.



How were they fine-tuned?

We ran up to 100 automated trials per model to find the exact configuration delivering the highest accuracy.



Final result

The blended approach consistently outperforms every individual method, even when tested on completely new customers.

AUC 0.82745



Take any two customers – one likely to leave and one likely to stay and we'll correctly identify who's who about 83 out of 100 times, even if they're completely new to us.



That 83% means that, in most cases, we're able to distinguish between at-risk and loyal customers far better than chance (which would be 50/50). It's a reliable signal you can use to spot risk early, focus retention efforts where they matter most, and protect revenue.

What 0.827 Actually Means for Us ?

5-7×



Cheaper to retain a customer than acquire a new one, every save compounds in ROI



Where We Are, and What Comes Next

What We'd Do With 6 More Months

We Know Who, Not Why

The model flags at-risk customers accurately but can't yet explain which product/experience triggered the risk. Front-line teams need a reason to act, not just a score.

Scores Decay Over Time

Customer behaviour shifts. A model trained today will lose accuracy within 6–12 months without retraining.
No retraining schedule exists yet.

No Feedback Loop

We don't know if outreach actually worked. Without tracking which flagged customers were retained, the model can never improve from real-world outcomes.

Recommended Next Steps

Run a 90-Day Pilot

Pick 1,000 flagged customers. Call half, leave half alone.
Measure the difference in retention. Prove the ROI before scaling.

Build the Feedback Loop

Connect model scores to CRM outcomes.
Every retained customer becomes a data point that makes the next model stronger.

Make It Speak Business Language

Work with the data team to translate anonymised features into plain terms - "low balance trend" instead of "saldo_var30."



The Business Opportunity

A Conservative Estimate: \$256M Revenue Protected

6.4M

At-risk customers
identifiable today

10%

Of those we conservatively
expect to save
through targeted outreach

640K

Customers retained
per year

\$256M

Revenue protected
(\$400 avg. annual
revenue per customer)

$$6.4M \times 10\% \text{ save rate} \times \$400 \text{ avg. revenue} = \$256M$$



**The customers most at risk
rarely say a word.
Now, we don't need them to.**



Thankyou!

